

# Coding-Decoding

A code is a type of signal. Coding is a method of transmitting a message between the sender and the receiver without a third person knowing it. Before transmitting the data is encoded and at receiver side encoded data is decoded in order to obtain original data by determining common key in encoded data.

## Approach to solve the questions of this section

- observe alphabets or numbers given in the code keenly.
- Find the sequence it follows whether it is ascending or descending.
- Detect the rule in which the alphabets/numbers/words follow.
- Fill the appropriate letter/number/word in the blank given.

## Types of Coding and Decoding

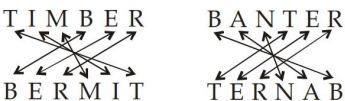
1. Letter coding
2. Number coding
3. Substitution
4. Mixed letter coding
5. Mixed Number coding

## Solved Examples

### Letter Coding:

1. If in any coded language TIMBER is written as BERMIT then in same coded language what would stand for BANTER?

- (a) TERNAB (b) RETNAB  
(c) TENBAR (d) TABNER

**Sol. (a);** 

### Number Coding:

2. If 7, 15, 15, 4 is GOOD and 2, 1, 20, 20, 5, 18 is BATTER, what number represent BEST?

- (a) 2, 5, 19, 20 (b) 2, 19, 4, 20  
(c) 2, 4, 19, 20 (d) 2, 5, 20, 19

**Sol. (a);** Letters are coded according to their place value in English alphabet, So, BEST @ 2, 5, 19, 20.

### Substitution:

3. If white is called blue, blue is called red, red is called yellow, yellow is called green, green is called black, black is called violet and violet is called orange, what would be the colour of human blood?

- (a) Red (b) Green  
(c) yellow (d) violet

**Sol. (c);** Yellow

The colour of human blood is 'red' and as it is given that 'red' is called 'yellow'. So, the colour of human blood is 'yellow'.

### Mixed letter coding:

4. In a certain code language:

- (1) 'pod na joc' means 'very bright boy';  
(2) 'tam nu pod' means 'the boy comes';  
(3) 'nu per ton' means 'keep the doll';  
(4) 'joc ton su' means 'very good doll'.

Which of the following means 'bright' in the same code language?

- (a) joc (b) pod  
(c) ton (d) na

**Sol. (d);** na

From statement 1 and 2, pod is common and the common word is 'boy'. So, 'Pod' stands for 'boy'.

From statement 1 and 4, joc is common and the common word is 'very'.

So 'joc' stands for 'very'.

So, 'na' stands for 'bright'.

### Mixed number coding:

5. In a certain code,

- '786' means 'study very hard'  
'958' means 'hard work pays'  
'645' means 'study and work'.

Which of the following is the code for 'very'?

- (a) 7 (b) 5  
(c) 8 (d) 6

**Sol. (a); 7**

In first and second statements, the common word is 'hard' and the common code digit is '8'. So, '8' means 'hard'.

In the first and third statements, the common word is 'study' and the common code digit is '6'. So, '6' means study. Thus, in the first statement '7' means 'very'.

### Practice Set

- In a certain code, BASIC is written as DDULE. How is LEADER written in that code?  
(a) NGCFGT (b) NHCGGU  
(c) OGDFHT (d) OHDGHU
- In a certain code, SPRING is written as UNUFRC. How is MOBILE Written in that code?  
(a) KQEFPA (b) OMDGNC  
(c) OMDGPA (d) OMEFPA
- In a certain code, POETRY is written as QONDSQX and OVER is written as PNUDQ. How is MORE written in that code?  
(a) LNNQD (b) NNNQD  
(c) NLNQD (d) NLPQD
- In a certain code, GERMINATION is written as IMGRENNOAIT. How is ESTABLISHED written in that code?  
(a) BEATSLEIHS (b) BAETSLEDIHS  
(c) BATESLDEIHS (d) BAETSLDEIHS
- In a certain code language, COMPUTER is written as RFUVQNPC. How is MEDICINE written in that code?  
(a) EOJDEJFM (b) EOJDJEFM  
(c) MFEDJJOE (d) MFEJDJOE
- If in a certain language, COVALENT is written as BWPDUOFM and FORM is written as PGNS. How is SILVER be written in that language?  
(a) MJTUDQ (b) KHRSFW  
(c) MJTWFS (d) MJTSFW
- In a certain code, COMPATIBLE is written as BQNPDDKAHS. How is STABILISED written in that code?  
(a) TUBCJCDRHK (b) JCBUTEDTHM  
(c) JCBUTCDDRHK (d) JCBUTEFTJM
- If in a certain language, CREATIVE is written as BDSBFUJS. How is TRIANGLE written in that language?  
(a) BHSSFKHM (b) BHSSMHHF  
(c) BSHSFHKM (d) BSHFMKH
- If in a certain language, CLOUD is written as GTRKF. How is SIGHT written in that language?  
(a) UGHHT (b) UHJFW  
(c) WFJGV (d) WGJHV
- If 'DELHI' is coded as '73541' and 'CALCUTTA' is coded as '82589662' then how 'CALICUT' will be written in that code language?  
(a) 5978213 (b) 5279431  
(c) 8251896 (d) 8543691
- In a certain code "CERTAIN" is written as "QDBVOJB". How is "RELATED" written in that Code.  
(a) QDKCVFE (b) KDQCEFU  
(c) DKCQEFV (d) KDQCVE
- In a certain code "MADRAS" is written as "NZEQBR" then "CALCUTTA" will be written as.  
(a) OZMBVSUB (b) BBKBTVSB  
(c) DZMBVUUZ (d) DZMBVSUZ
- In a certain code "BROWN" is written as "ZPMUL" then "VIOLET" will be written as.  
(a) TGMJCR (b) SGMTCQ  
(c) TGMJCQ (d) TGWCQ
- If "BRASS" is coded as "CTBUT", "AMIT" is coded as "BOJV" then what will be code of "ADITYA".  
(a) BEJUZZ (b) CEKVZB  
(c) BFJZVC (d) BFJVZC
- In a certain code "UNDER" is written as "6152@" and "DEAF" is written as "52#7". How "FRAUD" is written is that code.  
(a) 7@6#5 (b) 72#65  
(c) 7@#65 (d) None of these
- In a certain code "89654" is written as "MNOPQ" and "1634" is written as "KOLQ". How "PNKLQ" will be written in that code.  
(a) 59134 (b) 58134  
(c) 59234 (d) 59143
- In a certain code "TREAD" is written as "7%#94" and "PREY" is written as "\$%#8". How is "ARTERY" written in that code?  
(a) 9#7%#8 (b) 9#%7#8  
(c) 9%7%#8 (d) 9%#7%8
- In a certain code "SHINE" is code as "FOJIT" and "AFTER" is coded as "SFUGB". How will "PRITY" written in that code?  
(a) QSJUZ (b) ZUJSQ  
(c) ZUSJQ (d) ZQUSJ



**Directions (19-24):** Study the following information to answer the given question:

ADITYA is related to AYTIDA, SHUKLA is related to ALKUHS.

19. NUMBER is related to  
 (a) AEBMUN (b) REBMUN  
 (c) NUMBER (d) ABCDE
20. PROVIDE is related to  
 (a) EDIVORP (b) QSPWJEF  
 (c) FEJWPSQ (d) None of these
21. SANJAY is related to  
 (a) YJNAS (b) YBKOB T  
 (c) TBOKBZ (d) None of these
22. FIRST is related to  
 (a) TSRIG (b) JSTUJ  
 (c) MNOPQ (d) None of these
23. SAURABH is related to  
 (a) IBBSVBT (b) HBBRVAS  
 (c) HBASVBS (d) None of these
24. NIKHIL is related to  
 (a) LIHINK (b) LIHKIN  
 (c) LIHNIK (d) Cannot be determined

**Directions (25-30):** In a certain code "MANISH" is written as "NCOKTI", "MRADULA" is written as "NSCEWMC", "PINKI" is written "QKOLK".

25. What will be the code for "ANUP"  
 (a) COWQ (b) TQOCM  
 (c) CEKUZ (d) BOVQ
26. What will be the code for "SONAL".  
 (a) TPOBM (b) TQOCM  
 (c) TQOBM (d) TPOCM
27. What will be the code for "ADITYA"  
 (a) BEJUZB (b) CEKVAB  
 (c) CEKUZC (d) OWUCO
28. What will be the code for "NUTAN".  
 (a) OWUCO (b) PWVCP  
 (c) OWCUP (d) OWUCP
29. What will be the code for "MADHU".  
 (a) NCEIM (b) NCAIM  
 (c) NCEIW (d) NCWIEC
30. What will be the code for "MANOJ".  
 (a) NCORK (b) NCOQK  
 (c) NCOCRK (d) NCROCK

31. If 'RAM' is coded as '14', 'SHYAM' is coded as '30' then how would 'MOHAN' will be written in that code language?

(a) 20 (b) 24  
 (c) 36 (d) 37

32. If 'green' means 'white', 'white' means 'yellow', 'yellow' means 'blue', 'blue' means 'rosy', 'rosy' means 'black' then what will be the colour of milk?

(a) yellow (b) green  
 (c) blue (d) rosy

33. If 'clouds', means 'white', 'white' means 'rain', 'rain' means 'green', 'green' means 'air', 'air' means 'blue', 'blue' means 'water' then, where the birds fly?

(a) Green (b) Air  
 (c) White (d) Water

34. If 'Earth' is known as 'Water', 'Water' is known as 'Sky', 'Sky' is known as 'Tree', 'Tree' is known as 'Shadow', 'Shadow' is known as 'Fruit', 'Fruit' is known as 'Paper' the where will fish swim?

(a) Water (b) Sky  
 (c) Shadow (d) Paper

35. If in a code language 'finger' known as 'toe', 'toe' known as 'foot', 'foot' known as 'thumb', 'thumb' known as 'ankle', 'ankle' known as 'palm', 'palm' known as 'knee' then in that code language, what will an uneducated man use to mark his signature?

(a) Toe (b) Knee  
 (c) Thumb (d) Ankle

36. In a code language 'Square' is called 'Rectangle', 'Rectangle' is called 'Triangle', 'Triangle' is called 'Trapezium', 'Trapezium' is called 'cylinder', 'cylinder' is called 'cone', 'cone' is called 'circle', 'circle' is called 'pentagon' then the formula ' $\pi r^2 h$ ' used for?

(a) Cylinder (b) Cone  
 (c) Rectangle (d) Trapezium

37. In a certain code language '4×6' means '34', '9×6' means '69' then what would be the meaning of '7×7'?

(a) 49 (b) 56  
 (c) 63 (d) 64

38. If 'A' is coded as '4', 'B' is coded as '8', 'C' is coded as '5', 'O' is coded as '1', 'L' is coded as '6', 'N' is coded as '7', 'Q' is coded as '9'. To keep in mind above code what would be suitable alphabet in place of ?.

|   |   |   |   |   |
|---|---|---|---|---|
| A | Q | ? | B | B |
|---|---|---|---|---|

= 36

(a) B (b) L  
 (c) N (d) Q

39. If 'Green' is called 'Yellow' is called 'White', is called 'Red', 'Red' is called 'Rosy', 'Rosy' is called 'Black' then what will be the colour of human blood?

(a) Red (b) Black  
 (c) White (d) None of these

- Direction (41 to 45) :** Given below are some words in column I and their equivalents in column II. The letters in the codes are not necessarily in the same order as the letters in the words to which they stand but the same code stands for the same word. Study the two columns carefully and answer the questions.

41. What will be the code for word 'BIRDS'?  
(a) ga sa re ma pa                      (b) ma ga re sa pa  
(c) sa re ma pa ga                      (d) ni ma pa ga re
42. What will be the code for the word 'BREAD'?  
(a) sa ni pa ma ga                      (b) ga ni ga pa ma  
(c) sa ma ni da pa                      (d) ga pa ni re ga
43. What will be code for word 'DREADED'?  
(a) pa ma ni da pa ma pa  
(b) pa sa ni ga pa ni pa  
(c) pa ma ni sa pa ni pa  
(d) pa ma ni da pa ni pa
44. What will be code for word 'SERBIA'?  
(a) ga ni ma sa re da                      (b) ga ni ma pa re da  
(c) ga ma ni sa re da                      (d) pa ga re ma ni sa
45. What will be the code for the word 'RAISED'?  
(a) ma da re ga ni pa                      (b) ma ga re sa ni pa  
(c) ma da pa ga sa ni                      (d) sa re pa ni ga da

| Column-I | Column-II  |
|----------|------------|
| RED      | botany     |
| BELL     | whistle    |
| DRUME    | oprintment |
| NATRON   | lute       |
| SCIENCE  | stone      |
| STEATITE | bud        |

47. What will be the code for 'RED'?
- (a) botany                                      (b) whistle
- (c) opriment                                    (d) bud
48. Alphabets are to be coded as following?

What will be the code for 'FIGHT'?

- (a)  $\# \# \# \# \# >$  (b)  $\# \# \# \# \# <$   
(c)  $\# \# \# \# \# <$  (d)  $\# \# \# \# \# <$

- $\Delta = 2$ ,  $\square = 3$ ,  $\bigcirc = 4$ ,  $\triangleup = 10$ ,  $*$  = 6,  $\parallel = 7$ ,  $\square = 5$   
then choose the best option to complete the answer.

(a)  $\triangle$  (b)  $\square$   
(c)  $\square$  (d)  $\times$   
(e)  $\bigcirc$

- |          |   |   |   |   |   |   |   |   |   |
|----------|---|---|---|---|---|---|---|---|---|
| Alphabet | A | C | E | G | I | K | M | O | Q |
| Code     | # |   |   |   | > | < | ^ | ≡ | ∇ |

(a)  $| \equiv \wedge ||$                       (b)  $||| \wedge \equiv \equiv$   
(c)  $| \equiv \wedge \equiv$                       (d)  $\wedge < > \vee$

51. If in a certain code, RAIL is written as KCTN and SPEAK is written as CGRUM. How will AVOID be written in that code?

52. In a code language 'Noun' is called 'Pronoun', 'Pronoun' is called 'Article', 'Article' is called 'Preposition', 'Preposition' is called 'Conjunction', 'Conjunction' is called 'Adverb', 'Adverb' is called 'Verb', 'Verb' is called 'Adjective' then, In which can we use 'IN', 'IT', 'FOR'?

53. If 'Lead' means 'Stick', 'Stick' means 'Nib', 'Nib' means 'Niddle', 'Niddle' means 'Rope', 'Rope' means 'Thread' then what can we use in pen that help us to write?

54. In a coding language 'Hydrosphere' is known as 'Lithosphere', 'Lithosphere' is known as



'Troposphere', 'Troposphere' is known as 'Stratosphere', 'Stratosphere' is known as 'Biosphere', 'Biosphere' is known as Atmosphere then where will be  $O_3$ ?

- (a) Lithosphere (b) Biosphere  
(c) Stratosphere (d) Atmosphere

55. In a coded language 'M' is coded as '26', 'E' is coded as '10' R is coded as '36' then what would be the perfect code of 'GOLD' in the same coded language?

- (a) 67 (b) 76  
(c) 56 (d) 84

**Direction (56 to 60) :** For each of the question given below a group of digits followed by four combinations of letters/symbols numbered (1), (2), (3), (4). The letters are to be coded as per the scheme and conditions given below. The serial number of combination that correctly represents the group of letters is your answer.

Digits/symbol codes:- 5 2 3 % 6 @ © 9

Letters :- N B D E U J K I

Digits/symbol codes :- 4 \$ 1 \* 8 7 #

Letters :- A P M T R H F

**Conditions :-**

- (i) If the first digit is odd and the last digit is even number, their codes are to be interchanged.

(ii) If the first element is a symbol and last element is a number then both are coded as 'y'.

(iii) If the first element is a number and last element is a symbol, both are coded by 'z'.

(iv) If first and last both elements are even number then both are to be coded by the code of first element.

56. 4 3 # © 16

- (a) ADFKMU (b) ADFKMA  
(c) UDFKMU (d) UDFKMA

57. @ 9 2 © \$ 5

- (a) JIBKPN (b) YIBKPY  
(c) ZIBKPN (d) NIBKPN

58. 5 1 % 3 @ \$

- (a) NMEDJP (b) PMEDJN  
(c) YMEDJY (d) ZMEDJZ

59. 9 @ 2 4 6 3

- (a) IJBAUD (b) DJBAUI  
(c) IJBAUI (d) DJBASI

60. 2 # 4 1 9 7

- (a) B F A M I H (b) H F A M I B  
(c) B F A M I B (d) Z F A M I Z

### Previous year Questions (Memory Based)

1. If in a certain code, LUTE is written as MUTE and FATE is written as GATE, then how will BLUE be written in that code?

- (a) CLUE (b) GLUE  
(c) FLUE (d) SLUE

2. If in a certain language, MADRAS is coded as NBESBT, how BOMBAY coded in that language?

- (a) CPNCBX (b) CPNCBZ  
(c) CPOCBZ (d) CQOCBZ

3. If FISH is written as EHRG in a certain code, how would JUNGLE be written in that code?

- (a) ITMFKD (b) ITNFKD  
(c) KVOHMF (d) TIMFKD

4. In a certain code, TWINKLE is written as SVHOJKD then how would FILTERS be written in that code?

- (a) EHKSDQR (b) EHKUDQR  
(c) EGHUQDR (d) GJMSFST

5. In a certain code, ROAD is written as URDG. How is SWAN written in that code?

- (a) VXDQ (b) VZDQ  
(c) VZCP (d) UXDQ

6. If in a certain language, OPERATION is written as NODQBUJPO. How is INVISIBLE written in that language?

- (a) JOWJTJCMF (b) JOWJTHAKD  
(c) HMUHTJCMF (d) HMUHTHAKD

7. In a certain code, FAVOUR is written as EBUPTS. How is DANGER written in that code?

- (a) CBFFDS (b) CBMHDS  
(c) EBFHDS (d) EBHHFS

8. If SUMMER is coded as RUNNER, the code for WINTER will be

- (a) SUITER (b) VIOUER  
(c) WALKER (d) SUFFER

9. In a certain code, PRODUCTIONS is written as QQPCVEUHPMT. How is ORIENTATION written in that code?

- (a) PQJDOVBSJNO (b) PQJDOUBUJPO  
(c) PSJFOVBSJNO (d) NSHFMVBSJN

10. If in a code, MIND becomes KGLB and ARGUE becomes YPESC, then what will DIAGRAM be in that code?

- (a) BGYEPYK (b) BGYPYEK  
(c) GLPEYKB (d) LKBGYPK

11. If 'EARN' is written as 'RANE' in a certain code and 'BOND' is written as 'NODB' then how would 'TEAR' be written in the same code?  
 (a) EART (b) ATRE  
 (c) AERT (d) REAT
12. If 'PAGES' is written as 'RDIHU' in a certain code and 'WRITE' is written as 'YUKWG' then how would 'OTHER' be written in that same code?  
 (a) PWIHS (b) RVKGU  
 (c) RWJHU (d) QWJHT
13. If in a certain language 'SPORADIC' is coded as 'QNORDJEB' then how would 'TROUBLES' be coded in that same language?  
 (a) SQTNTFMC (b) TNQSRDKA  
 (c) TNQSTFMC (d) TFQSCMFT
14. If in a certain language 'LAWN' is coded as 'JCUP' then how would 'SLIT' be coded in that same language?  
 (a) QNGV (b) QJGV  
 (c) QNVG (d) NJGV
15. If cod for 'SATELLITE' is 'FUBTLDSHK', then what would be the code for 'LAUNCHING'?  
 (a) DOUBFGMHO (b) OVBMCFMHG  
 (c) OVMBFCFMHG (d) DOUBCFMHG
16. In a coded language 'FORMATION' is written as 'ZSXTJOBSL' and 'RACIAL' is written as 'XJNBQJ'. How will 'RATIONAL' be written in that same code language?  
 (a) XJOBSLJQ (b) JXOBSLJQ  
 (c) XJOBSJLQ (d) JXOBSJLQ
17. In a coding system, 'KINETIC' is written as 'TICDKIN'. How is 'MACHINE' written in that same code?  
 (a) ENIGMAC (b) INEGMAC  
 (c) INEGCAM (d) ENIGCAM
18. In a certain language 'DESCRIBE' is coded as 'FCJSDTFE', then how 'CONSIDER' coded in that same language?  
 (a) SFEJJOPED (b) SEFJTOPED  
 (c) QFETJOPD (d) None of these
19. If A = 1, BAN = 17, then INDIA = ?  
 (a) 37 (b) 36  
 (c) 35 (d) 9
20. If 'Y' is coded as 2, PEN is coded as = 11-22-13 then what word can we get from 10-6-18-24-16=?  
 (a) QUICK (b) QUITE  
 (c) JFRXP (d) QUACK

21. If 'GEAR' is coded as '5914' and 'ROUTE' is coded as '47289' then 'GATE' may be coded as  
 (a) 5187 (b) 5189  
 (c) 5289 (d) 5429
22. In a certain code, 'EAT' is written as '318' and 'CHAIR' is written as '24156', then how 'TEACHER' be written in that code language?  
 (a) 8313426 (b) 8312436  
 (c) 8321436 (d) 8312346
23. In a certain language, 'spread red carpet' means '247', 'dust one carpet' means '236', 'one red carpet' means '234'. Which of the following means 'dust' in that language?  
 (a) 2 (b) 3  
 (c) 6 (d) None of these

**Directions (24 - 27):** Here English letters are given from A to Z. Every capital letter is coded by a small letter.

|   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|
| A | B | C | D | E | F | G | H | I | J | K | L |
| j | n | l | r | i | x | a | v | e | s | o | y |

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| d | q | m | t | g | u | c | z | w | h | p | k | b | f |

Below there is a group of six letter as a question and given four column. a, b, c, d already coded. You should use these column as a answer. Read carefully the letters of group and give the answer.

| (a)    | (b)    | (c)    | (d)    |
|--------|--------|--------|--------|
| uhmvjr | nywgea | kegsor | vezuiv |
| nyweqa | tzeigj | tiqawe | keguqr |
| upfrvg | wqsjbl | bilpyq | wsqjbl |
| veziyu | upfurg | nywgca | kesgor |
| biqppu | uhmvjr | upfvra | blipoq |

24. BLUQSG  
 25. RWZHDG  
 26. HITREH  
 27. YCEWKN
28. In a certain code, 'DOWN' is written as '5 @ 9 #' and 'NAME' is written as '# 6 % 3'. How would 'MADE' be written in that code?  
 (a) %653 (b) % @ 6 3  
 (c) % 5 @ 3 (d) % @ 5 3
29. In a coding language, 'GUST' is coded as '@ 7 9 2', 'SNIP' is coded as '9 5 7 #', 'GAPE' is coded as '@ 5 3 5' then, What would be the code of 'SING'?  
 (a) 9 5 7 # (b) 5 9 # 5  
 (c) 9 B 7 5 (d) 9 7 5 @



**Direction (30 to 35) :** Read carefully the given information and choose the best option

In a certain code language 'her idea has merit' is written as 'for la bu na' and 'merit list has been displayed' is written as 'jo ke la si na' and 'her name displayed there' is written as 'ya si bu zo' and 'name in merit list' is written as 'na ya go ke'.

30. What is the code of 'ke'?  
(a) been (b) has  
(c) merit (d) list
31. What is the code of 'Idea'?  
(a) for (b) la  
(c) bu (d) na
32. What is the code of 'zo'?  
(a) there (b) displayed  
(c) name (d) her
33. What is the code of 'in'?  
(a) na (b) ya  
(c) go (d) ke
34. How can we coded 'her name is there'?  
(a) zo ya go wo (b) bu ya zo go  
(c) zo ya bu ke (d) ya zo wo bu
35. How can we code name has been displayed ?  
(a) ya la ke si (b) jo si ya la  
(c) si jo ke na (d) bu ya ke la

36. If the word 'CHRISMAS' written as 'HCIRMSSA' then how can we express the word 'CHRYSENT-HEMUMA' in the same language?  
(a) HCRYESTNEHUMMA  
(b) CHYRESTNEHUMAN  
(c) CHYRESTNEHMUAM  
(d) HCYRESTNEHUMAM
37. If in certain code, 'LANGUAGES' is coded as 'AGGNUAELS' then how is 'BUILDINGS' coded in the same language?  
(a) ILNDIUGBS (b) ILNIDUGBS  
(c) INLDIGUBS (d) ILNDIUGSB
38. What message we can get form given code?  
TCHLI USEIC SLAUS EECCL RUEET  
(a) CLUE is clear (b) This is a secret  
(c) Lies are classic (d) Use lesser chilli
39. If 'VIJAY' is written as 'DIAAG' in a certain code, then how would 'SURAJ' be written in that same code?  
(a) JCIAA (b) JCAIA  
(c) AJCIJ (d) JCIAJ
40. If 'RAHUL' is written as 'IZSFO', 'SAURAV' is written as 'HZFIZE'. How will 'SACHIN' be written in that same coded language?  
(a) HZXMRs (b) HZXSRM  
(c) HZACDP (d) AZOPQM

### Practice Set Solutions

1. (b); The letters at the odd-numbered positions in the word are each moved two steps forward while those at the even-numbered positions are each moved three steps forward to obtain the corresponding letters of the code.
2. (d); The first, third and fifth letters of the word are each moved two, three and four steps forward respectively while the second, fourth and sixth letters are moved two, three and four steps backward respectively to obtain the corresponding letters of the code.
3. (c); The first letter of the word is replaced by a set of two letters—one following it and the other preceding it—in the code. The remaining letters of the word are each moved one step backward to obtain the remaining letters of the code, in order.
4. (d); The middle letter remains unaltered in the code. Let us label the five letters before the middle letter as well as those after it, from 1 to 5. Then, the

code contains the letters of each group in the order, 5, 4, 1, 3, 2. Thus we have:

ESTABLISHED → BAETSLDEIHS  
1 2 3 4 5 1 2 3 4 5 5 4 1 3 2 5 4 1 3 2

5. (b); Each letter of the word except the first and last letters, is moved one step forward and then the order of the letter is obtained, is reversed to get the code.
6. (d); The letters in the first half and second half of the word are written in the reverse order and then each letter of the group so obtained is moved one step forward to get the code. Thus, we have:  
SILVER → SIL/VER → LIS/REV → MJT/SFW
7. (c); The letters in the first half and second half of the word are written in the reverse order and in the group of letters so obtained each letter in the first half is moved one step forward while in the second half is moved one step backward, to get the code. Thus, we have: STABILISED → STABI/

8. (a); The letters of the first half and the second half the words are written in the reverse order and then in the group of letters so obtained, the first, third, fifth and seventh letters are each moved one step forward while the second, fourth, sixth and eighth letters are each moved one step backward to get the code.

Thus, we have: TRIANGLE → TRIA/NGLE → AIRT/ELGN → BHSSFKHM

9. (d); The letters of the word are written in the reverse order and in the group of letters so obtained, the first, third, fifth letters are each moved three steps forward while the second and fourth letters are each moved one step backward to get the code.

10. (c); As; D - 7 and C - 8 similarly  
 E - 3 A - 2 C - 8  
 L - 5 L - 5 A - 2  
 H - 4 C - 8 L - 5  
 I - 1 U - 9 I - 1  
 T - 6 C - 8  
 T - 6 U - 9  
 A - 2 T - 6

11. (b);
- |   |    |   |   |    |   |
|---|----|---|---|----|---|
| C | -1 | Q | R | -1 | K |
| E |    | D | E |    | D |
| R |    | B | L |    | Q |
| T | +2 | V | A | +2 | C |
| A | +1 | O | T | +1 | E |
| I |    | J | E |    | F |
| N |    | B | D |    | U |

So, KDQCEFU

12. (d);
- |   |    |   |   |    |   |
|---|----|---|---|----|---|
| M | +1 | N | C | +1 | D |
| A | -1 | Z | A | -1 | Z |
| D | +1 | E | L | +1 | M |
| R | -1 | Q | C | -1 | B |
| A | +1 | B | U | +1 | V |
| S | -1 | R | T | -1 | S |
|   |    |   | T | +1 | U |
|   |    |   | A | -1 | Z |

So, DZMBVSUZ

13. (a); Each letter of the word BROWN has been coded two letters behind in the alphabetic order. So code of VIOLET is TGMJCR.

14. (d);
- |   |    |   |   |    |   |   |    |   |
|---|----|---|---|----|---|---|----|---|
| B | +1 | C | A | +1 | B | A | +1 | B |
| R | +2 | T | M | +2 | O | D | +2 | F |
| A | +1 | B | I | +1 | J | I | +1 | J |
| S | +2 | U | T | +2 | V | T | +2 | V |
| S | +1 | T |   |    |   | Y | +1 | Z |
|   |    |   |   |    |   | A | +2 | C |

15. (c); compare UNDER and DEAF you see that DE is common and in code "52" is common in same pattern you find that code as follows:

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| U | → | 6 | D | → | 5 | F | → | 7 |
| N | → | 1 | E | → | 2 | R | → | @ |
| D | → | 5 | A | → | # | A | → | # |
| E | → | 2 | F | → | 7 | U | → | 6 |
| R | → | @ |   |   |   | D | → | 5 |

16. (a); from given code we find that code of PNKLQ will be "59134"

17. (c);
- |   |   |   |   |   |    |
|---|---|---|---|---|----|
| T | → | 7 | P | → | \$ |
| R | → | % | R | → | %  |
| E | → | # | E | → | #  |
| A | → | 9 | Y | → | 8  |
| D | → | 4 |   |   |    |

So ARTERY is coded as 9%7#%8

18. (b); from reverse order each letter change to its next letter. so PRITY is coded ZUJSQ.

Logic (19-24): Each word of the series is rearrange in reverse order.

19. (b); REBMUN  
 20. (a); EDIVORP  
 21. (d); YAJNAS  
 22. (d); TSRIF  
 23. (d); HBARUAS  
 24. (b); LIHKIN

Logic (25-30): Each constant in series change to next letter while each vowel change to two letter next. So,

25. (a); COWQ  
 26. (b); TQOCM  
 27. (c); CEKUZC  
 28. (a); OWUCO  
 29. (c); NCEIW  
 30. (b); NCOQK



31. (b); As R A M  
18 1 13  
 $1+8+1+1+3=14$   
S H Y A M  
19 8 25 1 13  
 $1+9+8+2+5+1+1+3=30$   
M O H A N  
13 15 8 1 14  
 $1+3+1+5+8+1+1+4=24$

32. (b); There are 2 cases. Ist — the case of 'means' — in this we have to take answer as the backward part. IInd case — the case of known as, called as or written as — in this we have to take answer as the forward part. Therefore the answer this is green

33. (a); Green

34. (b); We know that fish swim in water but here water is called sky.

35. (d); We know that an uneducated man use thumb for signature but her thumb is called ankle.

36. (b); The formula ' $\pi r^2 h$ ' is used by 'cylinder' but here cylinder is called cone.

37. (c); As :-  $4 \times 6 = 24 + 4 + 6 = 34$   
and  $9 \times 6 = 54 + 9 + 6 = 69$   
Hence :  $7 \times 7 = 49 + 7 + 7 = 63$

38. (c); 

|   |   |   |   |   |
|---|---|---|---|---|
| A | Q | ? | B | B |
|---|---|---|---|---|

 = 36

$$4 + 9 + ? + 8 + 8 = 36$$

$$? + 29 = 36$$

$$? = 36 - 29$$

$$? = 7 \text{ is code for 'N'}$$

39. (d); The colour of human blood is Red. But here 'Red' is called Rosy.

40. (b); We use soap for washing clothes but here soap is called 'ink'.

**Directions (41 - 45) :**

From column I & II codes are as follows.

- I ® re  
S ® ga  
B ® sa  
R ® ma  
D ® pa  
E ® ni  
A ® da

41. (c); B I R D S  
— — — — —  
sa re ma pa ga

42. (c); B R E A D  
— — — — —  
sa ma ni da pa

43. (d); D R R A D E D  
— — — — —  
pa ma ni da pa ni pa

44. (a); S E R B I A  
— — — — —  
ga ni ma sa re da

45. (a); R A I S E D  
— — — — —  
ma da re ga ni pa

**Directions (46 - 47)**

46. (c); Codes are given on the basis of number of alphabets. Word 'SCIENCE' has seven alphabets.

so answer must be 'whistle'.

Whistle also have seven alphabets.

47. (d); Word 'RED' contains three alphabets so answer must be 'bud'.

48. (a); F I G H T  
↓ ↓ ↓ ↓ ↓  
# # # # >

49. (c);  $\Delta \times \square + \square - \triangle = 10$   
 $\Delta = 2, \square = 3, \square = 7, \triangle = 10$   
 $2 \times 3 + 7 - 10$   
 $= 6 + 7 - 10$   
 $= 13 - 10$   
 $= 3$

Hence :- 3 is coded by ' $\square$ '

50. (c); C O M A  
↓ ↓ ↓ ↓  
| # ^ #

### Distinct Solutions

51. (b); All the letters of the word, except the last letter, are written in the reverse order and in the group of letters so obtained, each letter is moved two steps forward to get the code. Thus, we have: AVOID → IOVAD → KQXCF

52. (d); We know in it, for are related preposition but here preposition is called conjunction.

53. (d); Stick.

54. (b);  $O_3$  is situated in Stratosphere but here Stratosphere is called Biosphere.

55. (b); As :-  $M = 13 \times 2 = 26$

$$E = 5 \times 2 = 10$$

$$R = 18 \times 2 = 36$$

$$\text{Then, } 7 - G \rightarrow 7 \times 2 = 14$$

$$15 - O \rightarrow 15 \times 2 = 30$$

$$12 - L \rightarrow 12 \times 2 = 24$$

$$4 - D \rightarrow 4 \times 2 = 8$$

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56. (b); 4 3 # © I 6  
↓ ↓ ↓ ↓ ↓ ↓

A D F K M A

(Condition (iv) is applicable)

57. (b); @ 9 2 © \$ 5  
↓ ↓ ↓ ↓ ↓ ↓

Y I B K P Y

(Condition (ii) is applicable)

58. (d); 5 1 % 3 @ \$  
- - - - -

Z M E D J Z

(Condition (iii) is applicable)

59. (a); 9 @ 2 4 6 3  
- - - - -

I J B A U D

(No condition is applicable)

60. (a); 2 # 4 1 9 7  
- - - - -

B F A M I H

(No condition is applicable)

### Previous year Solutions (Memory Based)

1. (a); The first letter of the word is moved one step forward to obtain the first letter of the code, while the other letters remain unaltered.

2. (b); Each letter in the word is moved one step forward to obtain the corresponding letter of the code.

3. (a); Each letter in the word is moved one step backward to obtain the corresponding letter of the code.

4. (b); Each letter in the word, except the middle letter, is moved one step backward while the middle letter is moved one step forward to obtain the corresponding letter of the code.

5. (b); Each letter in the word is moved three steps forward to obtain the corresponding letter of the code.

6. (c); Each of the first four letters in the word is moved one step backward, while each of the last five letters is moved one step forward to obtain the corresponding letter of the code.

7. (b); Each first, third and fifth letters are moved one step backward, while the second, fourth and sixth letters are moved one step forward to obtain the corresponding letter of the code.

8. (b); The first letter of the word is moved one step backward, while the two middle letters are each moved one step forward to obtain the corresponding letters of the code.

9. (a); The first, third, fifth, seventh, ninth and eleventh letters in the word are each moved one step forward; the second, fourth, eighth and tenth letters are each moved one step backward, while the middle (i.e. sixth) letter is moved two steps forward to obtain the corresponding letters of the code.

10. (a); Each letter in the word is moved two steps backward to obtain the corresponding letter of the code.

11. (c); similarly:-

12. (d); 

|        |        |                    |
|--------|--------|--------------------|
| P +2 R | W +2 Y | similarly:- O +2 Q |
| A +3 D | R +3 U | T +3 W             |
| G +2 I | I +2 K | H +2 J             |
| E +3 H | T +3 W | E +3 H             |
| S +2 U | E +2 G | R +2 T             |

13. (c); similarly:-



14. (a); L  $\xrightarrow{-2}$  J similarly:- S  $\xrightarrow{-2}$  O  
 A  $\xrightarrow{+2}$  C L  $\xrightarrow{+2}$  N  
 W  $\xrightarrow{-2}$  U I  $\xrightarrow{-2}$  G  
 N  $\xrightarrow{+2}$  P T  $\xrightarrow{+2}$  V

15. (b); S  $\xrightarrow{+1}$  F similarly:- L  $\xrightarrow{+1}$  O  
  
 L  $\xrightarrow{-1}$  D H  $\xrightarrow{-1}$  F

16. (a); F  $\rightarrow$  Z R  $\rightarrow$  X similarly:- R  $\rightarrow$  X  
 O  $\rightarrow$  S A  $\rightarrow$  J A  $\rightarrow$  J  
 R  $\rightarrow$  X C  $\rightarrow$  N T  $\rightarrow$  O  
 M  $\rightarrow$  T I  $\rightarrow$  B I  $\rightarrow$  B  
 A  $\rightarrow$  J A  $\rightarrow$  J O  $\rightarrow$  S  
 T  $\rightarrow$  O L  $\rightarrow$  Q N  $\rightarrow$  L  
 I  $\rightarrow$  B A  $\rightarrow$  J  
 O  $\rightarrow$  S L  $\rightarrow$  Q  
 N  $\rightarrow$  L

17. (b); K  $\rightarrow$  T similarly:- M  $\rightarrow$  I

18. (d); D E S C R I B E F C I S D T F E

Similarly: C O N S I D E R S F E I T O P D

19. (a); A = 1  
 B A N (2+1+14) = 17  
 I N D I A  $\rightarrow$  (9+14+4+9+1) = 37

20. (a); If Y = 2  
 P E N = 11-22-13  
 Then :- 10-6-18-24-16- Q U I C K  
 Codes are given in reverse sense.

21. (b); G - 5 R - 4 similarly  
 E - 9 O - 7 G - 5  
 A - 1 U - 2 A - 1  
 R - 4 T - 8 T - 8  
 E - 9 E - 9

22. (b); E - 3 C - 2 similarly  
 A - 1 H - 4 T - 8  
 T - 8 A - 1 E - 3  
 I - 5 A - 1  
 R - 6 C - 2  
 H - 4  
 E - 3  
 R - 6

23. (c); Spread red carpet  $\rightarrow$  247 ... (i)  
 Dust one carpet  $\rightarrow$  236 ... (ii)  
 One red carpet  $\rightarrow$  234 ... (iii)  
 from Eqs (ii) and (iii), One  $\rightarrow$  3  
 from Eqs (i) and (ii) and (iii), carpet - 2  
 Hence dust  $\rightarrow$  6

24. (c); From the given table it is obvious that code of BLUQSG  $\rightarrow$  nywgca which is present in option 3.

25. (c); Similarly the code of RWZHDG  $\rightarrow$  upfvra

26. (d); Similarly the code of HITREH  $\rightarrow$  vezuiv

27. (d); Similarly the code of YCEWKN  $\rightarrow$  blipoq

28. (a); D  $\rightarrow$  5 and N  $\rightarrow$  #  
 O  $\rightarrow$  @ A  $\rightarrow$  6  
 E  $\rightarrow$  9 M  $\rightarrow$  %  
 N  $\rightarrow$  # E  $\rightarrow$  3

similarly M  $\rightarrow$  %  
 A  $\rightarrow$  6  
 D  $\rightarrow$  5  
 E  $\rightarrow$  3

29. (d); G  $\rightarrow$  @ and S  $\rightarrow$  9 and G  $\rightarrow$  @  
 U  $\rightarrow$  7 N  $\rightarrow$  5 A  $\rightarrow$  5  
 S  $\rightarrow$  9 I  $\rightarrow$  7 P  $\rightarrow$  3  
 T  $\rightarrow$  2 P  $\rightarrow$  # E  $\rightarrow$  5

similarly S  $\rightarrow$  9  
 I  $\rightarrow$  7  
 N  $\rightarrow$  5  
 G  $\rightarrow$  @

(30 - 35):

Her idea has merit - for la bu na ... (i)  
 Merit list has been displayed - jo ke la si na ... (ii)  
 Her name displayed there - ya si bu zo ... (iii)  
 name in merit list - na ya go ke ... (iv)

30. (d); From Eqs (ii) and (iv) ke na → merit, list  
 From Eqs (iii) and (iv) and (i) na → merit  
 so ke → list

31. (a); From Eqs (i) and (iii) has → la  
 because (na for merit used above)

From Eqs (i) and (iii) Her → bu  
 so clearly Idea → for

32. (a); bu → her (from question 31)  
 from Eqs (iii) and (iv) → ya → name  
 from Eqs (ii) and (iv) → si → display  
 Hence : zo → there

33. (c); name → ya  
 merit → na  
 list → ke  
 so clearly → in → go

34. (d); Her name is there ?  
 Her → bu  
 name → ya  
 there → zo  
 is → the code of is totally different from the codes  
 given above

35. (b); name has been displayed ?  
 name → ya  
 displayed → si  
 has → la  
 been → jo

36. (d); As; C ↔ H, H ↔ C, R ↔ I, I ↔ R, S ↔ M, M ↔ S, A ↔ S, S ↔ A  
 similarly:- C ↔ H, H ↔ C, R ↔ Y, Y ↔ R, S ↔ E, E ↔ S, N ↔ T, T ↔ N, H ↔ E, E ↔ H, M ↔ U, U ↔ M, M ↔ A, A ↔ M

37. (b); LANGUAGES AGG NUAELS  
 1 2 3 4 5 6 7 8 9 6 4 7 3 5 2 8 1 9

Similarly,

BUILDINGS ILNIDUGBS  
 1 2 3 4 5 6 7 8 9 6 4 7 3 5 2 8 1 9

38. (b);

TCHLI USEIC SLAUSEECCL RUEET  
 THIS IS A SECRET

39. (a); (22) V  $\xrightarrow{(2+2=4)}$  D (4) similarly (19) S  $\xrightarrow{(1+9=10)}$  J (10)  
 (9) I  $\xrightarrow{(9=9)}$  I (9) (21) U  $\xrightarrow{(2+1=3)}$  C (3)  
 (10) J  $\xrightarrow{(1+0=1)}$  A (1) (18) R  $\xrightarrow{(1+8=9)}$  I (9)  
 (1) A  $\xrightarrow{(1=1)}$  A (1) (1) A  $\xrightarrow{(1=1)}$  A (1)  
 (25) Y  $\xrightarrow{(2+5=7)}$  G (7) (10) J  $\xrightarrow{(1+0=1)}$  A (1)

40. (b); As

Similarly,